Research Project and Methodologies (Biology L7)

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Research Project and Methodologies (Biology L7) (A36022)

Short Title:	Res Project & Meth (Biol L7)		
Faculty	Science and Computing		
Department	Science		
Cluster	Placement and Projects		
Author	NKENNEDY		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module consists of a research and development project, which will give students the facility to plan and carry out a body of work, based significantly on their own initiative. Students will also be introduced to the tools central to research methodologies including search strategies for accessing the scientific literature, appropriate referencing techniques and scientific writing skills.

Programmes

`RESA-0219`	BSc in Molecular Biology with Biopharmaceutical Science (`WD_SMBIO_D`)	3 / 6 / M

Indicative Content

- Projects will be based on areas such as: molecular biology, microbiology, enzymology, environmental analysis, bioremediation & biocatalysis and analytical procedures
- Students will search the scientific literature for research papers relevant to the project topic and produce a short background/introduction and discussion of project results in relation to these papers within their project report

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Plan, organise and carry out a basic research and development project.
2. Use library resources to search the scientific literature and databases relevant to the project.
3. Analyse data generated and compare to results from selected papers in the scientific literature.
4. Develop a personal work plan, including self-directed training on sustainability, and responsibility for own work.
5. Communicate project data in an oral format, using visual aids.
6. Complete a detailed written report, including a short review of relevant scientific literature, summary of the project, and relation of results to selected papers from scientific literature.

Learning and Teaching Methods

- Supervised project work: practical laboratory-based.
- IT tutorials: Use library resources to search the scientific literature and databases relevant to a chosen research topic.
- Consultations with project supervisor.
- Self-directed training on sustainability.

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	100%	1,2,3,4,5,6

Assessment Criteria

<40%: Inadequate achievement of the learning outcomes (insufficient level of research work performed and/or poor report writing and oral presentation skills suggesting a poor standard and an inadequate knowledge of the research topic at hand).

40%-49%: Will be able to show a basic level of achievement in relation to the key learning outcomes (an adequate level of research work performed together with adequate report writing and presentation skills).

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of some of the complexities of the topics and evaluate some results (a good level of research work performed together with good report writing and presentation skills).

60%-69%: In addition to the above, the student will demonstrate more detailed achievements of project goals and demonstrate an ability to evaluate knowledge acquired through a very high standard of research work performed supplemented with a high standard of report writing, placing results in context to scientific literature, and presentation skills.

70%-100%: All previous to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, predict general trends in behaviour, including exceptional behaviour, and critically evaluate and relate topics to scientific literature, demonstrated through an excellent standard of research work performed supplemented with a superior standard of report writing and presentation skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Tutorial	12	
Practical	48	
Independent Learning	75	

Essential Material

- "Ebrary." http://site.ebrary.com/lib/waterfordit/home.action
- "Moodle will be used to link students to websites, background details and submission deadlines etc." www.moodle.ie
- "SETU Library Catalogue." http://library.wit.ie/
- "SETU Library Databases." http://library.wit.ie/Resources/databases
- "United Nations - Sustainable Development Goals" https://sdgs.un.org/

Supplementary Material

- "N-TUTORR Digital Badge Course: Introduction to SDGs" https://mydigitalbackpack.ie/local/dashboard/home

Requested Resources

- Room Type: Biology Lab